The evolving Mobile Age

Modular cellphones, wearables and amping security on them. Phones seem to be in a state of constant evolution.

Update: Project Ara

Project Ara, Google's modular smartphone project, is expected to be available for purchase early next year. The phone is also expected to cost $50 (₹3,000 approx.) and will be available in plain grey colour in order to encourage buyers to customize it. All of this was revealed by Paul Eremenko, the team leader of Project Ara at its first developers conference at Google's Mountain View campus.

Project Ara deals with the development of a "modular" smartphone – a smartphone that a user can upgrade with individual components and parts instead of spending money on an entirely new phone. These components will be of the miniaturised variety and will also rely on 3D printing. Eremenko revealed that the chassis or frame of the Ara smartphone will last around five to six years and will hold all the various components in place with the help of electro-permanent magnets. The components will communicate with each other using the UniPro standard, a high-speed interface platform developed for use in mobiles.

Eremenko also weighed in on the fact that even though Project Ara supports Android, Android, as yet, does not support the drivers required for modular hardware. He said that because Google owned the project, that problem should be solved soon.

Project Ara –The world's first modular cell phone

Eremenko also revealed the expected timeline for Project Ara that includes two more developer conferences in July and September of this year respectively. The alpha build of Project Ara is expected in August of this year and the beta build in January 2015, which will be made available for purchase.

A Module Developers Kit for Project Ara has also been unveiled by Google. The Kit has been released by Google's Advanced Technology and Projects (ATAP). Mobile developers who were anxiously waiting for Google to release more details about the project will now get a better idea of what's the possibilities are with the Ara components. Project Ara is basically a concept of building a customized smartphone with modular components with the user deciding which components he/she would want (just like building an assembled PC). Google hung on to Motorola Mobility's ATAP division after selling Motorola to Lenovo.

Contents of the MDK:

It details how Google is considering different sets of configurations for different sized metal endoskeletons. The endoskeleton is basically a single board which will hold all the components of the phone together.

The Project Ara smartphone will come in three different sizes: a mini, medium and large (In future) sizes.

Google is currently working on different design of rib crosses on the endoskeleton.

Google notes that Ara smartphones cannot have vertical "spines or vertically configured modules for front panelling. All front-facing modules must run horizontal and won't have an end to lean on and will therefore use EPM and a ball-spring plunger assembly to ensure they remain in place.

The EPMs (Electro-Permanent Magnets) which keeps the modules of the Ara phone in place will have "two selectable states: the attach state and release state, corresponding to high and low levels of magnetic force."

Recently, Phonebloks had uploaded a video which revealed the team working on Project Ara (http://dgit.in/pbltara) The MDK of Project Ara can be downloaded from http://dgit.in/mdkara.

Ara devices will most likely be compatible with most standard apps and ATAP has encouraged developers "to pay attention to application stability and graceful behavior in light dynamically varying availability of hardware resources (as with hot-plug, for instance)."

The iWatch (and other wearables)

According to reports, Apple has received samples of flexible PCBs from leading Flex-